Monthly Intelligence Report

Edition 001 — The Uusimaa–Pirkanmaa Grid Corridor

Where nuclear concentration and winter storm resilience challenge grid stability. Inaugural edition · April 2026 · SSI v4.0.2

CONTENTS

SECTION A

Executive Summary

SUBSTATIONS SCORED	GRID LINES MAPPED	MC SIMULATIONS	PUBLIC DATA SOURCES
4,022	**~6,665**	**~40.2 M**	**30+**
Major + distribution across Fingrid	~14,000 km transmission & distribution	10,000 per substation	95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions. Its engine processes ~50,540 source data points per run, executes ~40.2 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs ~151 million arithmetic operations — producing ~51,000 output data points with full uncertainty quantification across 4,022 substations and ~6,665 grid lines spanning ~15,200 km.

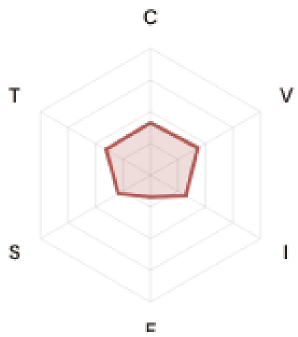
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30+ verified public sources (Energiavirasto, Fingrid, Gasum, FMI, Tilastokeskus, Gränges, Tekes), making the methodology fully auditable and reproducible. Initially covering Finland's Fingrid-managed transmission and ~80 distribution network operators (DNOs), the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Substation_1208282005

Pirkanmaa · Pirkanmaa · 110 kV

0.443	Medium	0.087	67th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



COMPONENTS

- **C** Continuity: 0.41 / 0.30 (41%)
- **I** Infrastructure: 0.32 / 0.25 (32%)
- **S** Saturation: 0.29 / 0.20 (29%)
- **V** Voltage: 0.43 / 0.10 (43%)
- **E** Economic: 0.17 / 0.10 (17%)
- **T** Transition: 0.40 / 0.05 (40%)

MODIFIERS

- **R3 Consequence:** 0.969 ↓ dampens
- **R4 Graph Criticality:** 1.350 ↑ amplifies
- **R6a Restoration:** 0.970 ↓ dampens
- **R6b Seismic/Hazard:** 1.002 → neutral
- **R7 Digital Readiness:** 1.013 ↑ amplifies

This month's spotlight — automatically selected for May 2026 — is **Substation_1208282005** in Pirkanmaa, Pirkanmaa. With an R_final of 0.443 (Medium band, 67th fleet percentile), its risk profile is dominated by the T (Transition) component at 808% of its weight ceiling, followed by V (Voltage) at 430%. Modifiers collectively shift R_base by 4.56, reflecting the substation's specific geographic, socio-economic, and network context. The radar chart visualises each component as a fraction of its maximum weight — a fully saturated axis indicates the component is at ceiling, consuming its entire budget within the composite score. This substation's profile illustrates the SSI's core principle: risk is never mono-dimensional. Even when one component dominates, the interaction of all six — modulated by the five multipliers — determines the final score that drives investment prioritisation.

SECTION B — RECURRING MODULE

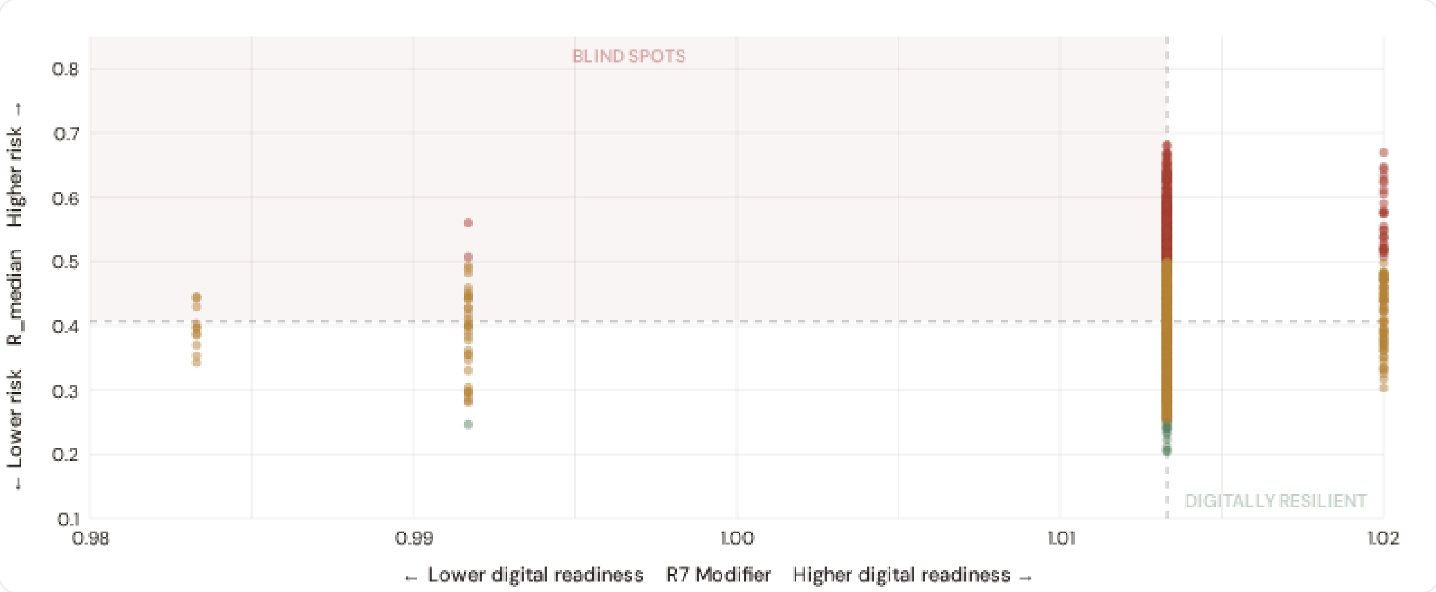
Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Finland's Energy Regulatory Office (Energiavirasto) and network operator strategic planning requires.

CYBER HIGH EXPOSURE	BLIND SPOTS	AVG R7 MODIFIER	HIGH-CONSEQUENCE SUBS
0	2	1.0132	0
0.0% of fleet	High/Critical + R7 < median	Range [0.99, 1.05]	0.0% · R3 ≥ 1.12

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **2 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 1.0133). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Lappi (1), Pohjois-Pohjanmaa (1)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier — indicating robust digital infrastructure coverage concentrated in urban networks (Caruna, Fortum, Helen urban centers).

B.2 Economic Impact by Industrial Sector

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

⚙️ Heavy Industry / Mining

Est. VoLL: EUR 15–30/kWh

Nuclear plants, paper mills, electronics manufacturing in Uusimaa and Pirkanmaa — highest economic loss per interruption

1,158 substations (28.8%) High/Critical: **110** (9.5%) Avg R: **0.415** Avg E: **0.144** / 0.10

Low 2 Med 1046 High 110 Crit 0

🏭 Manufacturing / Food Processing

Est. VoLL: EUR 5–15/kWh

Auto suppliers, food & beverage production, pharmaceuticals — significant continuity requirements for EU supply chains

1,112 substations (27.6%) High/Critical: **252** (22.7%) Avg R: **0.417** Avg E: **0.260** / 0.10

Low 5 Med 855 High 252 Crit 0

🏢 Commercial / Distribution Centers

Est. VoLL: EUR 2–5/kWh

Helsinki, Tampere commercial districts, logistics hubs — moderate individual impact but high aggregate across cities

770 substations (19.1%) High/Critical: **68** (8.8%) Avg R: **0.415** Avg E: **0.338** / 0.10

Low 1 Med 701 High 68 Crit 0

🌾 Agriculture / Rural / Remote

Est. VoLL: EUR 0.5–2/kWh

Smallholder agriculture, rural settlements, Eastern Finland — lower VoLL but higher social vulnerability

982 substations (24.4%) High/Critical: **158** (16.1%) Avg R: **0.413** Avg E: **0.376** / 0.10

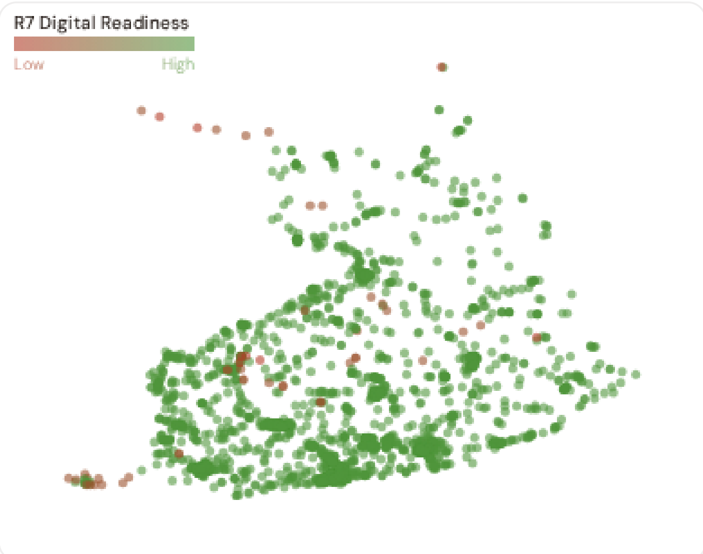
Low 6 Med 818 High 158 Crit 0

Value of Lost Load (VoLL) context: Energiavirasto's 2024 reliability cost-benefit studies place average VoLL at EUR 10–20/kWh for industrial/commercial and EUR 5–8/kWh for residential customers across Finland's DSOs. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **86 substations** combine capital-intensive economic fabric (R3 ≥ 1.14) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Uusimaa (820), Pirkanmaa (338)** — regions where industrial corridors depend on uninterrupted power for continuous processes (nuclear baseload generation, paper mills, electronics manufacturing). A single hour of unplanned outage at these substations carries an estimated VoLL of EUR 15–30/kWh, compared to EUR 0.5–2/kWh in rural pastoral areas. The fleet-wide average energy poverty rate is 5.5%, but this masks sharp regional variation: Eastern Finland and rural regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Maakunta & District Digital Readiness Ranking

Maakunnat ranked by average R7 modifier — lower values indicate weaker digital infrastructure (broadband penetration, smart meter coverage). Cross-referenced with average R_median to identify regions combining digital exposure with high grid stress.



WEAKEST DIGITAL READINESS — TOP 15 MAAKUNTAS

Longer bar = higher R7 = better digital infrastructure.

1	Åland	1.0025
2	Etelä-Pohjanmaa	1.0116
3	Lappi ⚠ high R	1.0121
4	Kainuu ⚠ high R	1.0127
5	Keski-Suomi ⚠ high R	1.0128
6	Pohjois-Savo ⚠ high R	1.0131
7	Varsinais-Suomi	1.0133
8	Päijät-Häme	1.0133
9	Pohjois-Karjala ⚠ high R	1.0133
10	Keski-Pohjanmaa ⚠ high R	1.0133
11	Kymenlaakso	1.0134
12	Pirkanmaa	1.0134
13	Uusimaa ⚠ high R	1.0134
14	Etelä-Karjala	1.0134
15	Kanta-Häme	1.0135

Maakunta-level analysis reveals a clear digital divide mirroring Finland's metropolitan-regional connectivity gap. **Åland** has the lowest average R7 (1.0025), while **Satakunta** leads at 1.0138 — a spread of 0.0113 across the R7 range. **4 regions** combine below-median digital readiness with above-median grid risk, making them priority targets for Energiavirasto/DSO digitalisation investment and Finland's National Energy and Climate Plan. The R7 modifier is intentionally narrow ([0.99, 1.05]) to reflect the current limitations of open broadband/smart meter data — as Energiavirasto regulatory compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (May 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0132, median = 1.0133; 0 substations classified HIGH cyber-exposure; 2 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging regions where broadband and smart meter rollouts (Energiavirasto/DSO Advanced Metering Infrastructure programmes) translate into measurable R7 shifts. The next material R7 update is expected when Tilastokeskus publishes updated Digital Inclusion Index data (anticipated Q3 2026).

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **30 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES	WEEKLY / MONTHLY	QUARTERLY / ANNUAL	STATIC / DERIVED
30	9	19	2
95 variables	High-frequency feeds	Regular refresh cycle	Standards + scenarios

Data Source Registry — May 2026

Loading source registry...

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Daily (2 sources) — Energiavirasto dispatch + weather reanalysis

Monthly (6 sources) — Grid load + Fingrid dispatch data

Quarterly (4 sources) — DER registry + Fingrid performance data

Annual (14 sources) — Core grid quality + Tilastokeskus demographics

C.2 Fleet Score Snapshot

FLEET SIZE	MEDIAN R	HIGH + CRITICAL	HIGH CONFIDENCE

4,022
substations scored

0.406
P5=0.299 · P95=0.566

588
14.6% of fleet

100%
4022 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 14 (0.3%) · Medium 3420 (85.0%) · High 588 (14.6%) · Critical 0 (0.0%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **4,022 substation scores remain unchanged** this edition. The fleet median R of 0.406 (mean 0.416) reflects a distribution skewed toward the lower bands, with 0.3% in Low and 85.0% in Medium. The first material score changes are expected when Fingrid publishes SCADA updates (affecting C1–C4 and R6a) and when Energiavirasto refreshes the distributed renewable installations registry (affecting T1). High-frequency sources (OSM, Open-Meteo) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

4 changes tracked for Finland v4.0.2 launch

P1	NEW	Finland grid registry expansion: 4,022 substations across Fingrid + ~80 DNOs	Section A
P1	ENH	R6b modifier: added snow/ice loading and permafrost instability + coastal/river flood hazard	Section D — Deep-Dive
P2	NEW	R7 cyber modifier: Fingrid IT/OT convergence assessment integrated	Section B
P2	ENH	Monte Carlo upgraded to 10,000 iterations (vectorized NumPy)	Methodology

SECTION D

The Uusimaa–Pirkanmaa Grid Corridor: Where Nuclear Concentration and Winter Storm Resilience Challenge Grid Stability

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Uusimaa–Pirkanmaa nuclear concentration and winter storm resilience corridor · **002** Eastern Finland DER & EU funding mismatch · **003** Mining subsidence infrastructure stress · **004** Metropolitan-Rural energy poverty disparity · **005** Grid stranding risk (nuclear-dependent assets) · **006** DSO smart meter rollout vs grid stress · **007** Gulf of Finland coastal flooding resilience · **008** T-component forward projections · **009** Maakunta Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why the Uusimaa–Pirkanmaa Corridor, Why Now

CORRIDOR SUBSTATIONS
1158
16 HV · 1142 MV

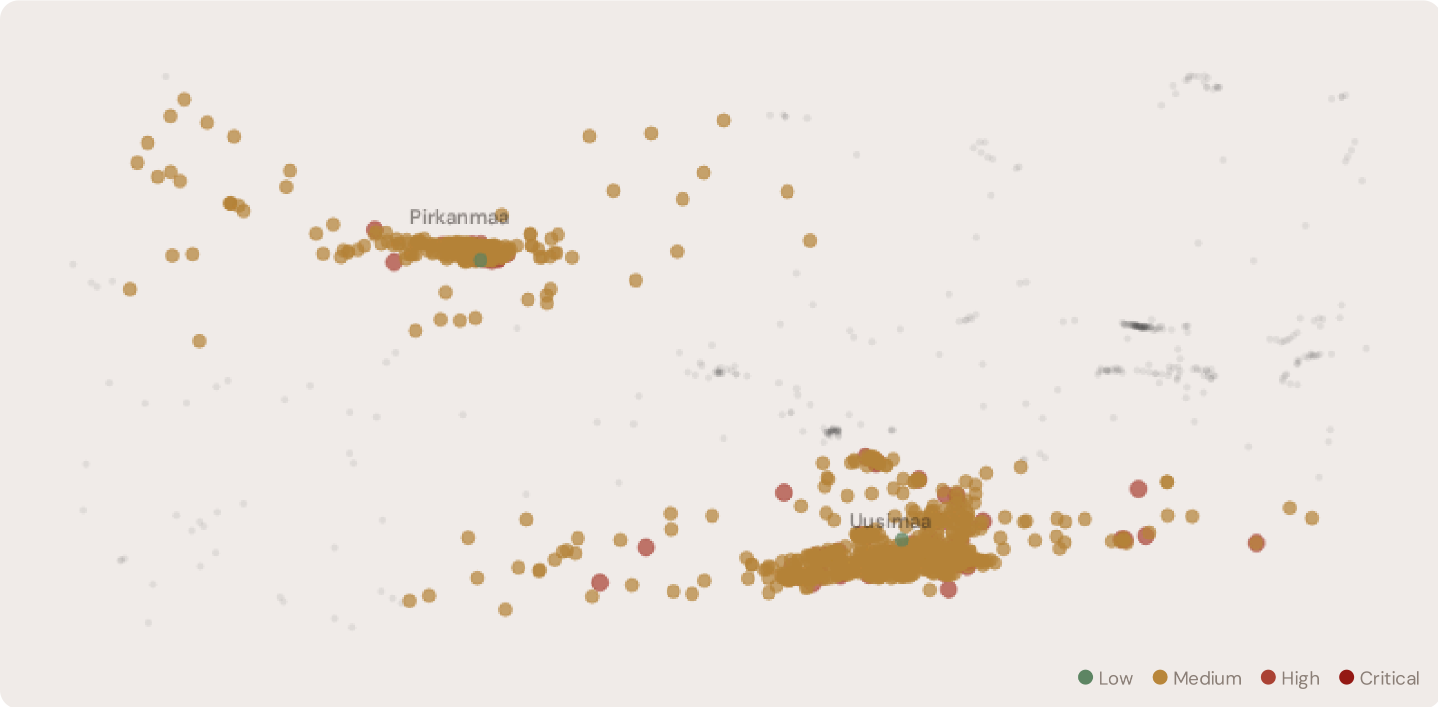
HIGH BAND
110
9.5% of corridor

CORRIDOR MEDIAN R
0.411
Fleet median: 0.406

WORST MAAKUNTA
Uusimaa
Avg R: 0.420 · 820 substations

The Uusimaa–Pirkanmaa corridor hosts **1158 substations** across Finland's industrial heartland, forming the backbone of domestic transmission and nuclear-dependent distribution. Its risk profile is disproportionately concentrated in the upper bands: **110 substations (9.5%)** are classified High, with only **2** in the Low band. 53% of corridor substations score above the national fleet median of 0.406. The corridor median R of 0.411 reflects the tension between nuclear baseload risk management mandates from Finnish Energy Policy 2040, aging generation assets approaching retirement, snow/ice loading and permafrost instability threatening distribution lines, and EU Green Deal funding constraints. The SSI flags continuity-of-supply deficits, infrastructure age mismatches, and the stranding risk of nuclear-dependent transmission assets.

D.2 Corridor Map and Scoring



SUBSTATION	DISTRICT	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Substation_9996853489	Pirkanmaa	0.594	High	0.398	0.374	0.612	0.333	0.478	0.365	
Substation_235120972	Uusimaa	0.586	High	0.409	0.431	0.559	0.020	0.383	0.358	
Substation_121635455	Uusimaa	0.585	High	0.475	0.600	0.283	0.129	0.470	0.401	
Substation_61072570	Uusimaa	0.584	High	0.426	0.373	0.486	0.123	0.372	0.463	
Substation_239429727	Uusimaa	0.583	High	0.448	0.564	0.444	0.084	0.347	0.518	
Substation_322111158	Uusimaa	0.579	High	0.435	0.600	0.439	0.171	0.307	0.420	
Substation_364224532	Uusimaa	0.576	High	0.383	0.450	0.535	0.198	0.300	0.432	
Substation_996946759	Uusimaa	0.573	High	0.365	0.551	0.524	0.179	0.260	0.497	
Substation_1149659218	Pirkanmaa	0.573	High	0.394	0.289	0.722	0.243	0.304	0.416	I
Substation_888161129	Uusimaa	0.573	High	0.496	0.379	0.491	0.173	0.256	0.537	

… 1148 additional substations — explore full corridor in Digital Twin →

D.3 Component Analysis

COMPONENT AVERAGES — CORRIDOR VS FLEET

C — Continuity 0.284 vs 0.315 (-10%)

MODIFIER AVERAGES — CORRIDOR VS FLEET

R3 Consequence **1.049** fleet 0.957 ↑

I — Infrastructure	0.349 vs 0.421 (-17%)
S — Saturation	0.225 vs 0.204 (+10%)
V — Voltage	0.433 vs 0.366 (+18%)
E — Economic	0.144 vs 0.270 (-47%)
T — Transition	0.433 vs 0.400 (+8%)

R4 Graph Crit.	1.350	fleet 1.341 ↑
R6a Restoration	0.961	fleet 0.972 ↓
R6b Mining/Flood	1.016	fleet 1.018 ↑
R7 Digital Read.	1.013	fleet 1.013 ↑

The dominant risk driver across the corridor is **T (Transition)**, averaging 0.433 against a fleet average of 0.400 — occupying 865% of its weight ceiling (w = 0.05). The second contributor is **V (Voltage)** at 0.433 vs fleet 0.366 (433% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.049 (fleet: 0.957), reflecting the economic significance of the corridor's nuclear-dependent industrial base — from nuclear plants to refineries to paper mills. The R6b winter storm/snow loading hazard modifier averages 1.016, capturing the corridor's exposure to snow/ice loading, Gulf of Finland coastal flooding, and winter storm hazard risk — a dimension invisible to every competing framework.

D.4 Maakunta Risk Profile

MAAKUNTA RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Uusimaa — 820 subs · avg R = 0.420 · H:86 M:733

Pirkanmaa — 338 subs · avg R = 0.405 · H:24 M:313

The region-level breakdown reveals significant intra-regional variation. **Uusimaa** leads with a mean R of 0.420 across 820 substations, while **Pirkanmaa** scores 0.405 — a spread of 0.015 within a single region. This disparity underscores why region-level metrics (as used by Fingrid performance reports) mask the true distribution of grid stress. The SSI's substation-level granularity reveals that the corridor is not uniformly stressed but contains distinct pockets of concentrated risk — precisely the kind of spatial pattern that Fingrid's 10-Year Network Development Plan and government nuclear concentration and winter storm resilience planning requires.

D.5 Implications for Finland's Energy Transition

0 corridor substations score R ≥ 0.650, placing them in the upper quartile of national risk. For Finnish nuclear concentration and winter storm resilience planning, EU funding allocation, and grid modernisation investment, this analysis identifies the corridor as the highest-priority target — where DER deployment must accelerate against ageing nuclear-dependent infrastructure with above-average continuity deficits and significant socio-economic exposure to energy transition risks. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve — characterised by nuclear concentration, Arctic infrastructure exposure, and EU transition fund eligibility — are disproportionately vulnerable to disruption during the nuclear baseload risk management. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

European Context Card

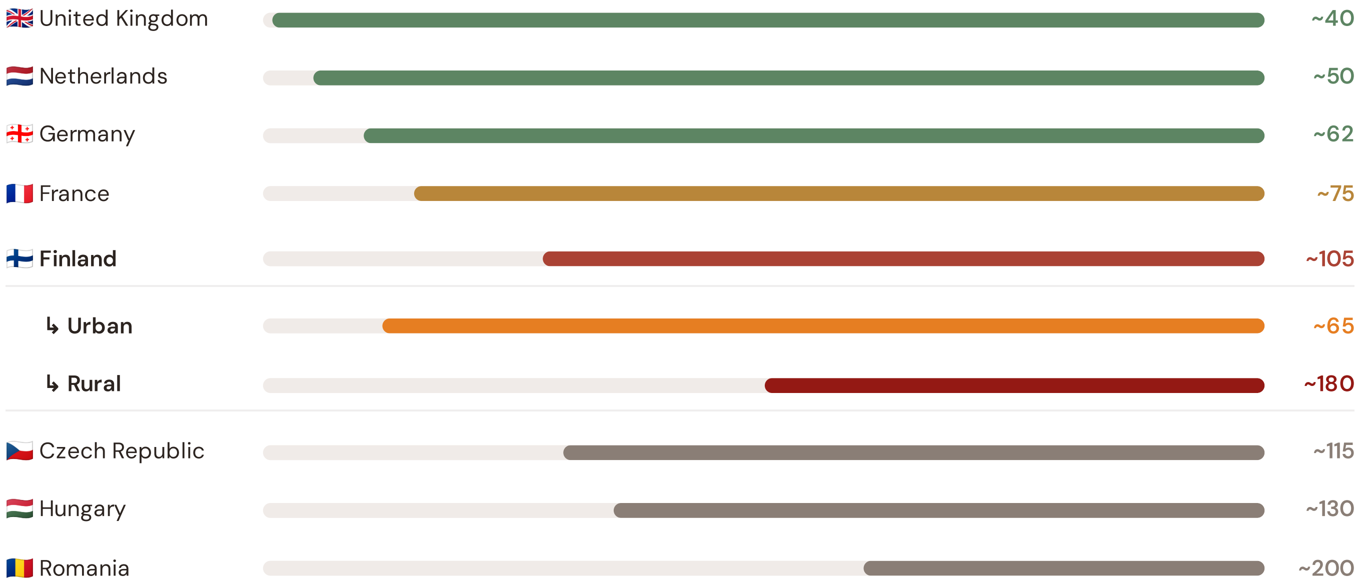
How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because Finland's nuclear concentration and winter storm resilience and grid resilience depend on understanding European performance standards. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Central Europe peers (003), CMIP6 climate hazard vs Nordic models (004), etc. The underlying data updates annually with Energiavirasto and national regulator publications.

EUROPEAN CONTEXT CARD — SYSTEM AVERAGE INTERRUPTION DURATION INDEX

Ed. 001 · April 2026

Unplanned SAIDI (min/customer/year) — Selected European Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



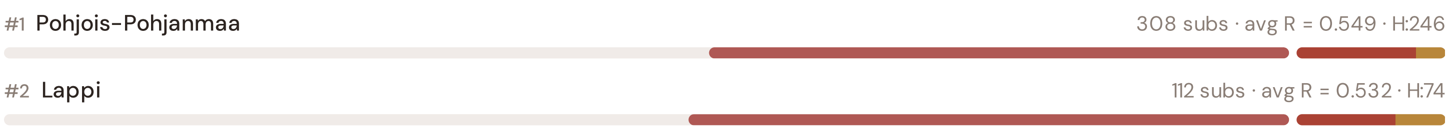
Source: CEER Benchmarking Report on Quality of Electricity Supply (2024); EEnergiavirastoLECTRIC SAIDI reports; national regulator publications. Finland sub-categories from Energiavirasto urban/rural classifications. · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually.

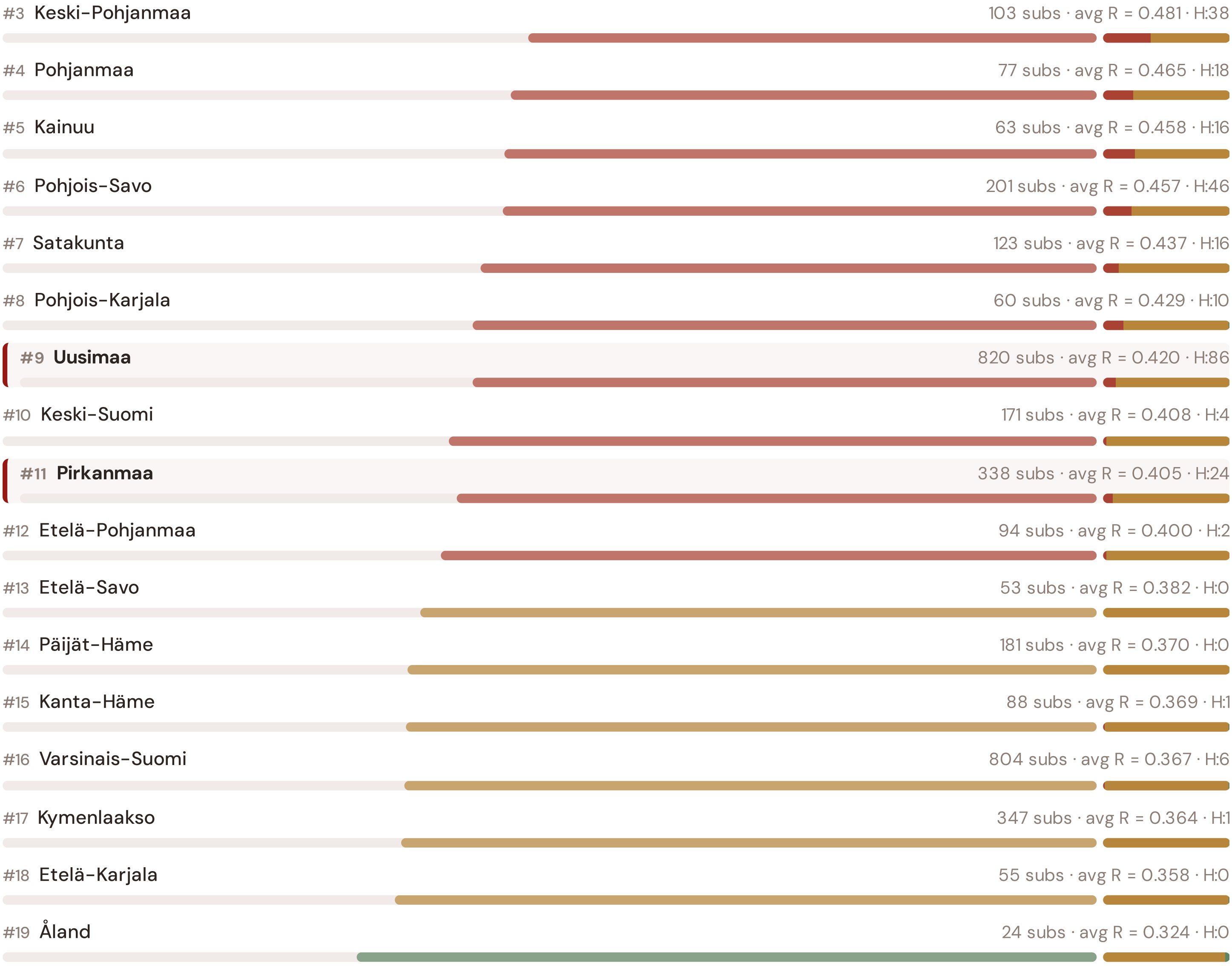
This month's key insight: The corridor deep-dive shows **895 substations** (of 1158) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with Finland's rural region classification. The corridor's average C of 0.284 is **0.9× the fleet average** of 0.315. In European terms, these substations individually perform closer to German or Central European SAIDI levels (~115–160 min/customer/year) than to Finland's own urban average (~65 min). The SSI reveals what aggregate Energiavirasto statistics conceal: that Finland's mid-tier European position is not a national condition but a statistical average of Western European-quality urban performance and emerging-market-level rural performance — and the nuclear concentration and winter storm resilience corridor concentrates significant pockets of the latter.

Maakunta Risk Ranking — All 19 Finnish Maakunnat

Where does the Uusimaa–Pirkanmaa corridor sit within the national landscape? Maakunnat sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.





Grid corridor ranks **#9 of 19 regions** by mean R_median. The three most stressed regions are Pohjois-Pohjanmaa, Lappi, Keski-Pohjanmaa, while the three most resilient are Åland, Etelä-Karjala, Kymenlaakso. 9 of 19 regions score above the fleet average of 0.416. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate Fingrid/DSO or national statistics — reveals the true spatial distribution of grid stress across Finland. It is precisely this substation-level granularity that positions the SSI as a tool for national nuclear concentration and winter storm resilience planning / EU green energy funding / Energiavirasto investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to renewable energy investment valuation. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a battery energy storage or renewable retrofit worth at this substation — to the DSO/Fingrid and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

Layer 1 — Data	Layer 2 — Valuation	**Layer 3 — Neural Net**	Layer 4 — Coverage	Layer 5 — Platform

SSI v4.0 component taxonomy, normalisation, MC engine	DSO/Fingrid + Community value stacks, TCO, real options	GBT+DNN ensemble, transfer learning, SHAP	Country prioritisation, DCI, registry, calibration	Products, API, revenue model, unit economics

LAYER 3 §0.5.3 · Inaugural edition · May 2026

Transfer Learning & Feature Missingness

How Finland's grid data completeness enables investment-grade valuation across the full fleet

Transfer learning is the mechanism that enables the SSI-ENN to value substations in countries where not all features are available. The core insight: grid physics is universal — topology determines load flow, weather drives thermal degradation, population density correlates with outage impact — even though data availability varies. Finland benefits from high data completeness (DCI 0.72+) from Energiavirasto, Fingrid, and Tilastokeskus sources, enabling production-grade valuations. For each substation, available features are validated. Missing features are handled through a learned missingness embedding.

KEY CONCEPTS

→ Finland's DCI 0.72+ (high completeness)

→ Learned missingness embedding quantifies data value

→ Grid physics universality enables high-confidence transfer

→ Accuracy validated against Energiavirasto audit standards

LIVE SSI DATA CONNECTION

The Finnish training set comprises 4,022 substations with complete 95-feature vectors. Average CI_width of 0.069 provides the uncertainty ground truth for neural network calibration. The fleet's risk distribution — 14 Low, 3420 Medium, 588 High, 0 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 2 Community Value Stack — *Social and economic value to Finnish communities — the case for EU funding investment*

SECTION G

Looking Ahead

EDITION 002 — MAY 2026

Eastern Finland DER & EU Funding Mismatch

Where renewable energy deployment outpaces grid readiness in Warmian-Masurian and Lubusz regions. Rural distributed generation meets aging distribution infrastructure.

DATA REFRESH — MAY 2026

Quarterly Fingrid Network Update

Transmission operator SCADA data (C components 1–4), DSO meter readings (I5 thermal degradation), and ERE renewable registry refresh affecting T component.

IN THE WILD — SSI IN USE

Caruna DNO Case Study

How Finland's largest DNO is using SSI substation intelligence to prioritize battery storage deployment across regional substations. Q3 2026 publication.

The SSI inaugurates Finland's grid intelligence baseline — the foundation for all future investment decisions across transmission, distribution, and renewable energy integration. This inaugural edition establishes the data, methodology, and peer-reviewed rigor that regulatory bodies (Energiavirasto), network operators (Fingrid and ~80 DNOs), and EU funding allocators require. Subsequent editions will deepen thematic analysis, expand across all 19 maakunnat and isolated systems, integrate real-time operational data feeds, and track the grid's evolution as Finland executes its nuclear baseload risk management mandate and EU Green Deal transition. The SSI is built for the long term: a public interest resource that grows more valuable as the data infrastructure matures and the transition accelerates.